

4. The number of digits in the square root of a perfect square number of n digits is

(i) $\frac{n}{2}$, if n is even

(ii) $\frac{n+1}{2}$, if n is odd

Here, $n = 12$. So, required number of digits = $\frac{n}{2} = \frac{12}{2} = 6$.

5. $\sqrt{\sqrt{17956} + \sqrt{24025}} = \sqrt{134 + 155} = \sqrt{289} = 17$.

6. $\sqrt{\sqrt{44944} + \sqrt{52441}} = \sqrt{212 + 229} = \sqrt{441} = 21$.

7. Sum of prime numbers greater than 4 but less than 16 = $(5 + 7 + 11 + 13) = 36$.

$\therefore \frac{1}{4} \times 36 = 9 = 3^2$.

8. Given exp. $= \sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + 15}}}}$
 $= \sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{169}}}}$
 $= \sqrt{10 + \sqrt{25 + \sqrt{108 + 13}}}$
 $= \sqrt{10 + \sqrt{25 + \sqrt{121}}}$
 $= \sqrt{10 + \sqrt{25 + 11}} = \sqrt{10 + \sqrt{36}}$
 $= \sqrt{10 + 6} = \sqrt{16} = 4$.

9. Given exp. $= \sqrt{41 - \sqrt{21 + \sqrt{19 - 3}}}$
 $= \sqrt{41 - \sqrt{21 + \sqrt{16}}} = \sqrt{41 - \sqrt{21 + 4}}$
 $= \sqrt{41 - \sqrt{25}} = \sqrt{41 - 5} = \sqrt{36} = 6$.

10. Given exp. $= \sqrt{176 + 49} = \sqrt{225} = 15$.

11. Given exp. $= \frac{14}{7} \times \frac{21}{7} \times \frac{120}{15} = 2 \times 3 \times 8 = 48$.

12. Given exp. $= \left(\frac{\sqrt{225}}{\sqrt{729}} - \frac{\sqrt{25}}{\sqrt{144}} \right) \div \frac{\sqrt{16}}{\sqrt{81}}$
 $= \left(\frac{15}{27} - \frac{5}{12} \right) \div \frac{4}{9} = \left(\frac{15}{108} \times \frac{9}{4} \right) = \frac{5}{16}$.

13. Let $(15)^2 + (18)^2 - 20 = \sqrt{x}$.

Then, $\sqrt{x} = 225 + 324 - 20 = 529$

$\Leftrightarrow x = (529)^2 = 279841$.

14. Let $\sqrt{x} \times \sqrt{484} = 1034$. Then, $\sqrt{x} \times 22 = 1034$

$\Leftrightarrow \sqrt{x} = \frac{1034}{22} = 47$

$\Leftrightarrow x = (47)^2 = 2209$.

15. Let $\sqrt{11881} \times \sqrt{x} = 10137$. Then, $109 \times \sqrt{x} = 10137$

$\Leftrightarrow \sqrt{x} = \frac{10137}{109} = 93$

$\Leftrightarrow x = (93)^2 = 8649$.

16. $\frac{4050}{\sqrt{x}} = 450 \Leftrightarrow \sqrt{x} = \frac{4050}{450} = 9$

$\Leftrightarrow x = 9^2 = 81$.

17. Let $\sqrt{\frac{16}{25}} \times \sqrt{\frac{x}{25}} \times \frac{16}{25} = \frac{256}{625}$.

Then, $\frac{4}{5} \times \frac{\sqrt{x}}{2} \times \frac{16}{25} = \frac{256}{625}$

$\Leftrightarrow \frac{64\sqrt{x}}{625} = \frac{256}{625}$

$\Leftrightarrow \sqrt{x} = \frac{256}{625} \times \frac{625}{64}$

$\Leftrightarrow x = 4^2 = 16$.

18. $\sqrt{(272)^2 - (128)^2} = \sqrt{(272 + 128)(272 - 128)}$
 $= \sqrt{400 \times 144} = \sqrt{57600} = 240$.

19. $6 * 24 = 6 + 24 + \sqrt{6 \times 24} = 30 + \sqrt{144}$
 $= 30 + 12 = 42$.

20. $10y\sqrt{y^3 - y^2} = 10 \times 5 \sqrt{5^3 - 5^2}$
 $= 50 \times \sqrt{125 - 25} = 50 \times \sqrt{100}$
 $= 50 \times 10 = 500$.

21. $\sqrt{110\frac{1}{4}} = \sqrt{\frac{441}{4}} = \frac{\sqrt{441}}{\sqrt{4}} = \frac{21}{2} = 10.5$.

22. $\sqrt{\frac{25}{81} - \frac{1}{9}} = \sqrt{\frac{25-9}{81}} = \sqrt{\frac{16}{81}} = \frac{\sqrt{16}}{\sqrt{81}} = \frac{4}{9}$.

23. Let $[(\sqrt{81})^2]^2 = x^2$. Then, $x^2 = (81)^2$ or $x = 81$.

24.

1	1	58	76	(126
22	58			
246	14	76		
	14	76		
	×			

 $\therefore \sqrt{15876} = 126$.

25.

1	3	00	00	00	(1.732
27	2	00			
343	11	00			
3492	71	00			
	69	84			
	1	16			

 $\therefore \sqrt{3} = 1.73 = \frac{173}{100}$.

26. A number ending in 8 can never be a perfect square.

27. The squares of numbers having 1 and 9 as the unit's digit end in the digit 1.

Such numbers are: 1, 9, 11, 19, 21, 29, 31, 39, 41, 49 i.e., there are 10 such numbers.